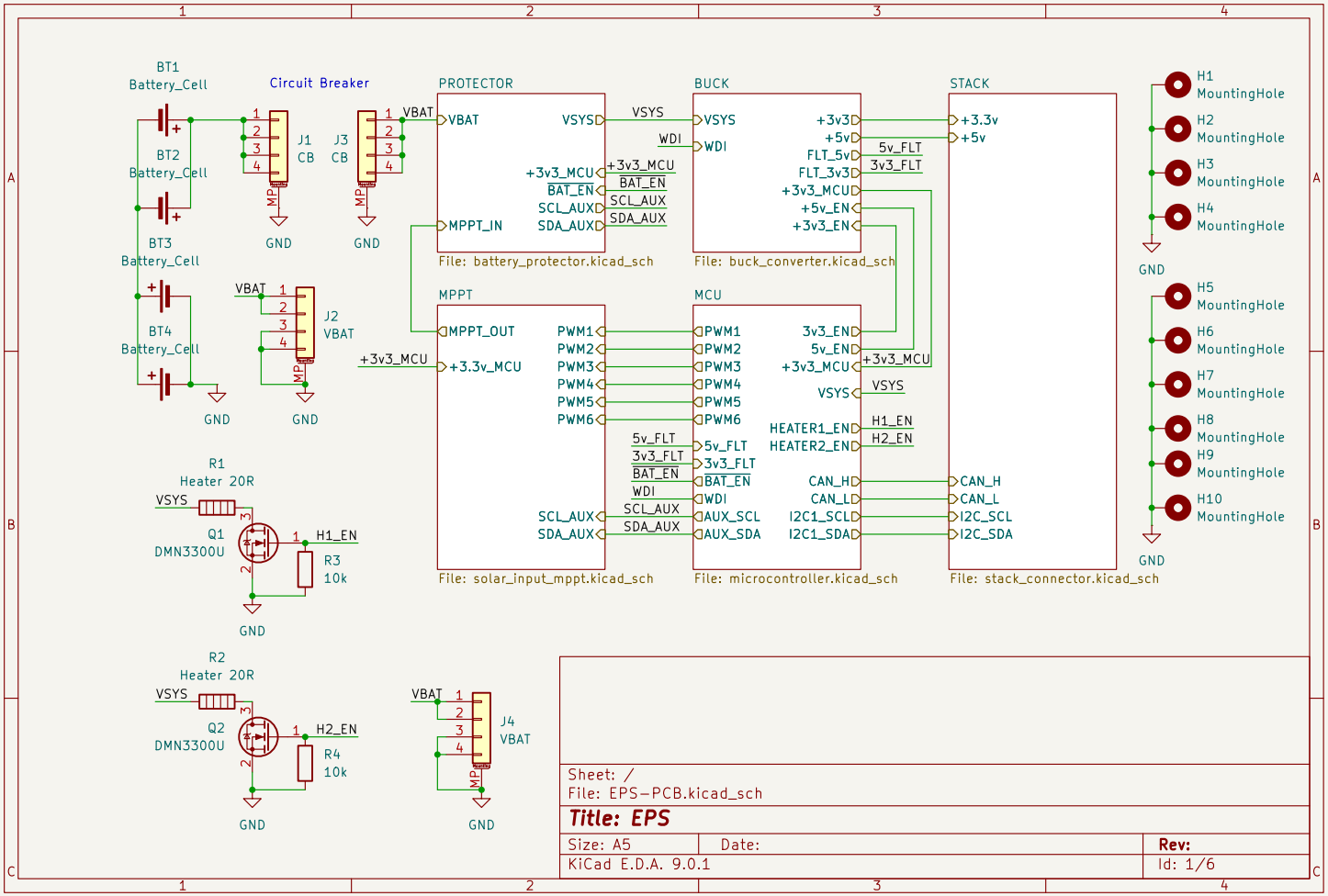
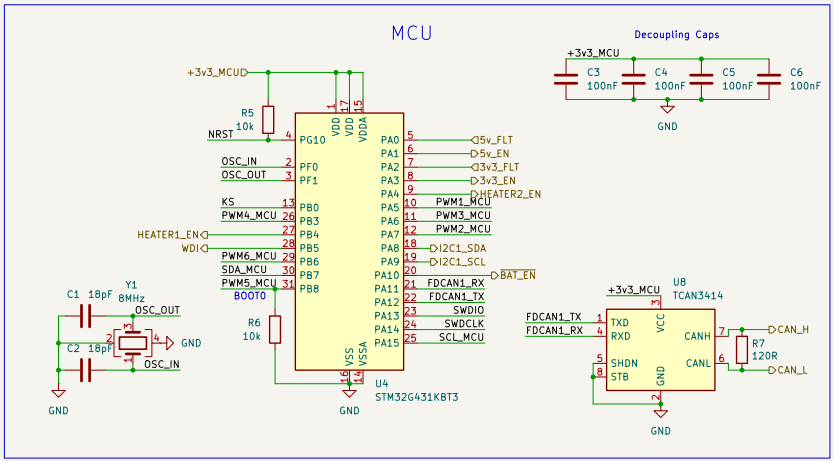
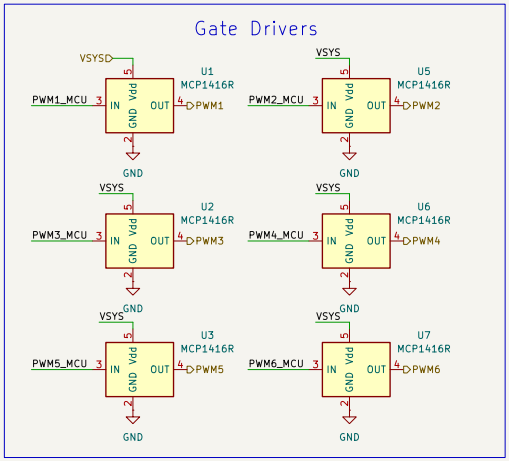


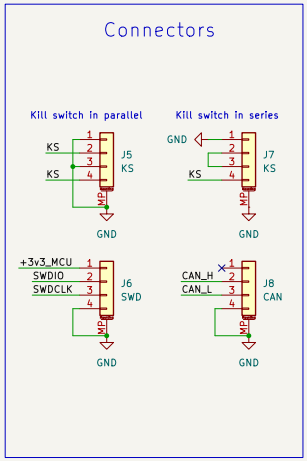
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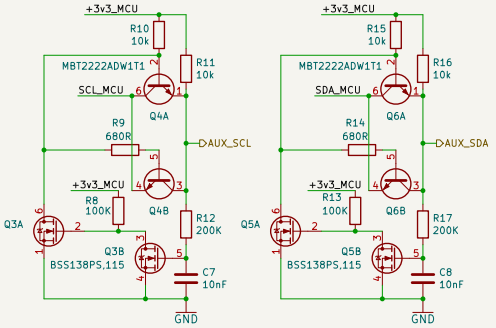


NOTE

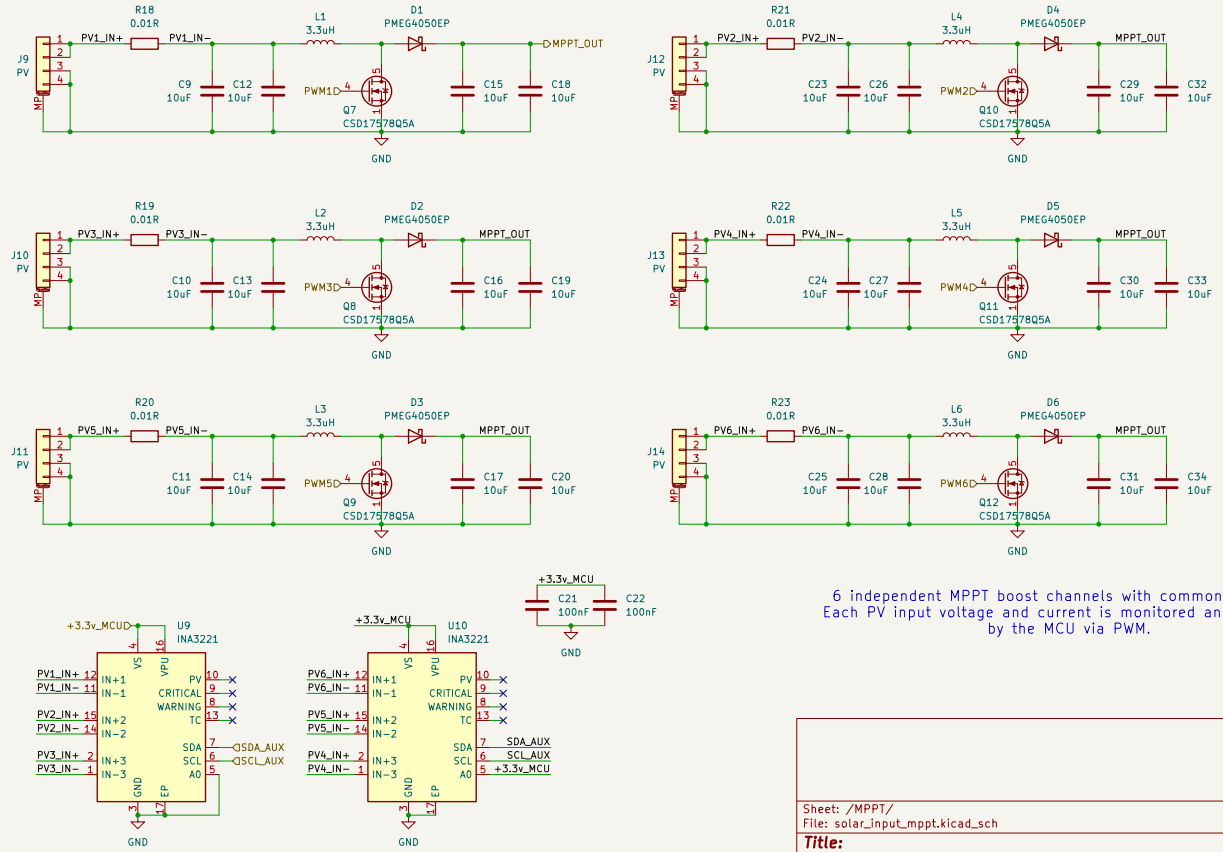
These novel bus protection circuits from PyCubed prevent traditional I2C/SPI failure modes where a single slave failure can disable the entire bus.

Learn more:
https://doi.org/10.36227/techrxiv.15166620

By default, slave clock and/or data lines can be held low and the Master (SAMD51) will still be able to communicate with the remainder of the bus.



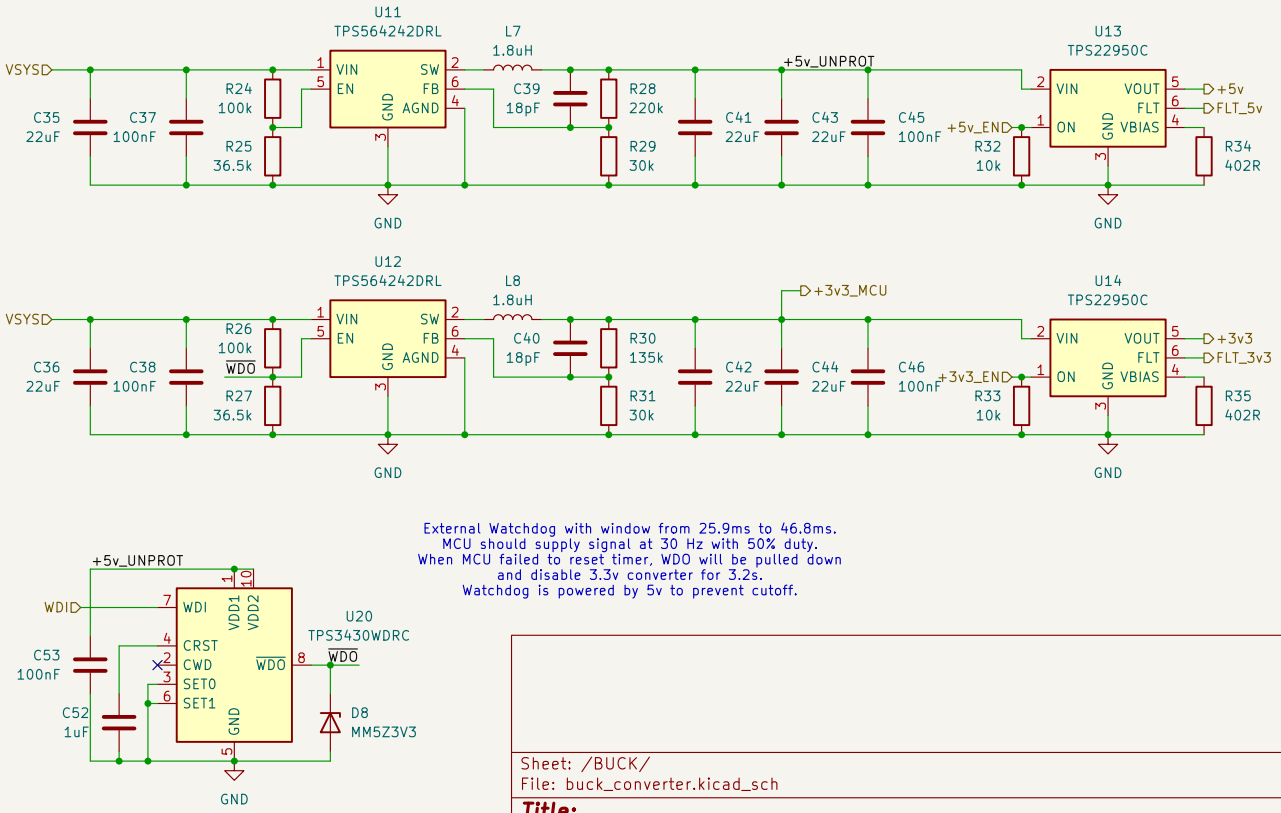
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Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.1		Id: 2/6



Sheet: /MPPT/
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Title:

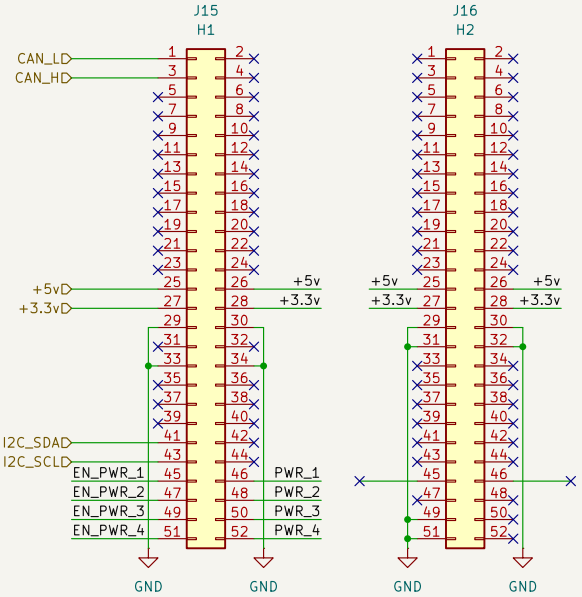
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KiCad E.D.A. 9.0.1		Id: 3/6



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File: buck_converter.kicad_sch

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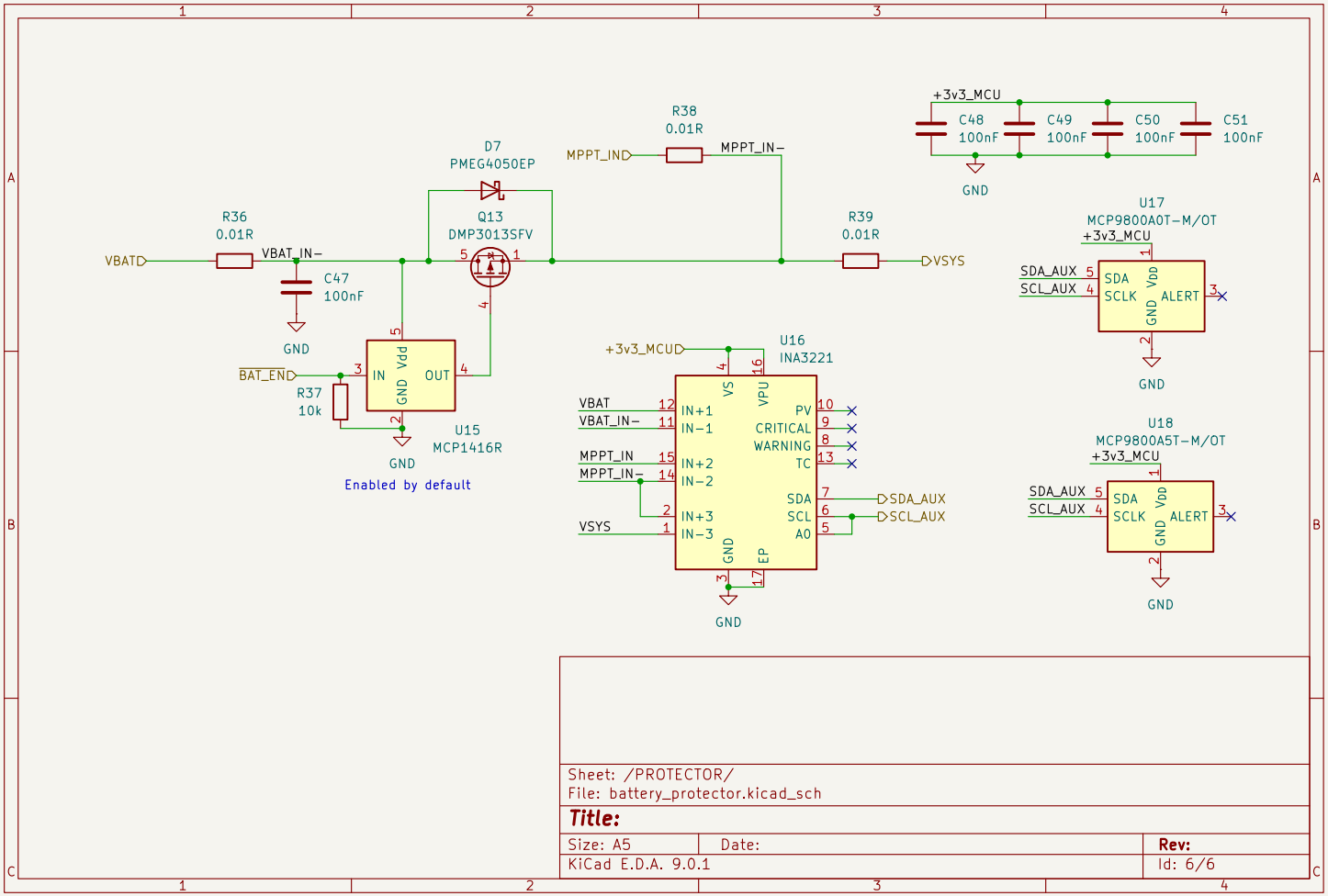
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KiCad E.D.A. 9.0.1		Id: 4/6



Sheet: /STACK/
File: stack_connector.kicad_sch

Title:

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KiCad E.D.A. 9.0.1		Id: 5/6


+3v3_MCU
C48
100nF
C49
100nF
C50
100nF
C51
100nF
GND
R38
0.01R
MPPT_IN
MPPT_IN−
D7
PMEG4050EP
Q13
DMP3013SFV
R36
0.01R
VBAT
VBAT_IN−
C47
100nF
GND
R39
0.01R
VSYS
U17
MCP9800A0T−M/OT
+3v3_MCU
SDA_AUX
SCL_AUX
SDA
SCLK
Vdd
ALERT
GND
BAT_EN
R37
10k
IN
GND
Vdd
OUT
U15
MCP1416R
GND
Enabled by default
+3v3_MCU
U16
INA3221
VBAT
VBAT_IN−
MPPT_IN
MPPT_IN−
VSYS
IN+1
IN−1
IN+2
IN−2
IN+3
IN−3
VS
VPU
GND
EP
PV
CRITICAL
WARNING
TC
SDA
SCL
A0
SDA_AUX
SCL_AUX
GND
U18
MCP9800A5T−M/OT
+3v3_MCU
SDA_AUX
SCL_AUX
SDA
SCLK
Vdd
ALERT
GND
GND
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Rev:
KiCad E.D.A. 9.0.1
Id: 6/6